Power

69 parts

File: power.kicad_sch

MCU

30 parts

File: mcu.kicad_sch

Sensors

24 parts

File: sensors.kicad_sch

Storage + USB

16 parts

File: storage_usb.kicad_sch

CAN

18 parts

File: can.kicad_sch

Connectors

46 parts

File: connectors.kicad_sch

GPS-denied quadrotor flight controller, rev B (DRAFT UNTIL INDEPENDENT REVIEW)

MCU: STM32H743VIT6, 480 MHz Cortex-M7, PX4 (NuttX). Pin map derived from Pixhawk FMUv6C.
IMUs: ICM-45686 (SPI1, external 32 kHz clock) + BMI088 (SPI2). Baro: BMP581 (I2C4). IMU heater on an isolated island (rev B).
Storage: microSD (SDMMC2) for logs, params and the RAPTOR policy.
Buses: 2x CAN FD (TCAN1462V), TELEM1/TELEM2 with flow control, GPS/I2C, RC (CRSF), USB-C.
Motors: M1-M4 on TIM1 (bidirectional DShot) via 8-pin ESC harness with telemetry and current; M5-M8 on AUX.
Power: 2S-6S in, reverse-polarity FET, TVS, 5 V 3.5 A buck, ideal-diode USB backup, 3V3 buck, low-noise sensor LDO, current-limited peripheral and CAN power.
Board: 38 x 38 mm, 30.5 mm M3 mount, 6-layer JLCPCB.

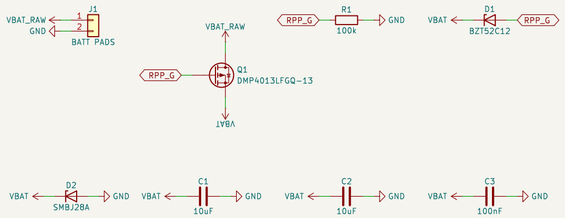
Generated from design.py; edit the source, not this file
Pin map: PX4 FMUv6C (Pixhawk 6C) derivative, STM32H743VIT6 LQFP-100
DRAFT UNTIL INDEPENDENT REVIEW. Not for fabrication.
Daniel K (dvntekim)

Sheet: /
File: fc.kicad_sch

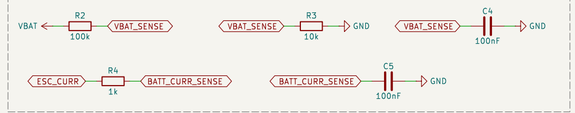
Title: GPS-denied quadrotor flight controller: Overview

Size: A3	Date: 2026-11-07	**Rev: B**
KiCad E.D.A. 9.0.8		Id: 1/7

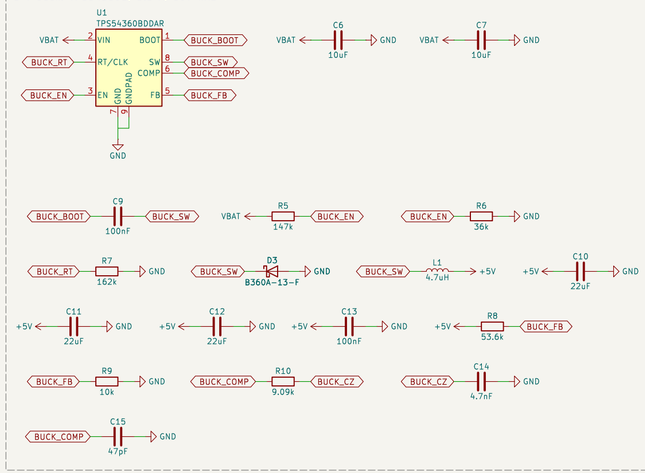
Battery input, reverse-polarity and surge protection



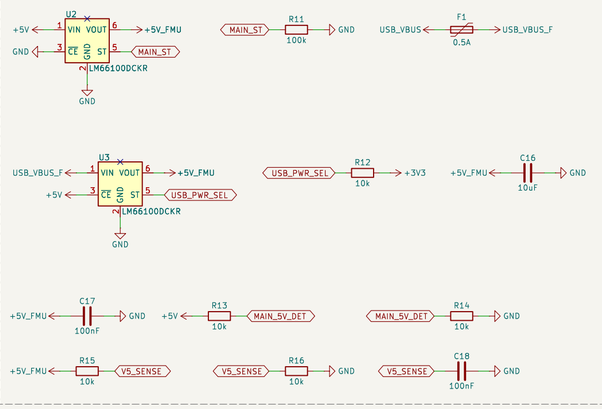
Battery voltage and ESC current sensing



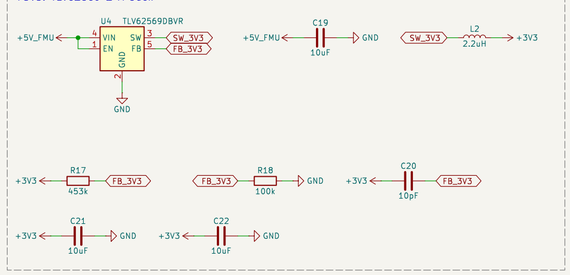
5 V buck: TPS54360B, 3.5 A, 597 kHz



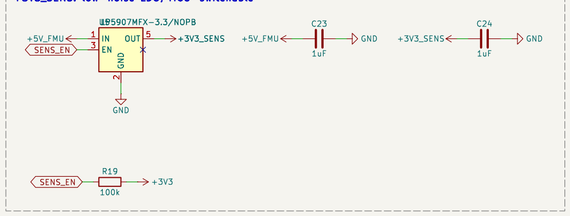
+5V_FMU: ideal-diode OR, buck has priority over USB



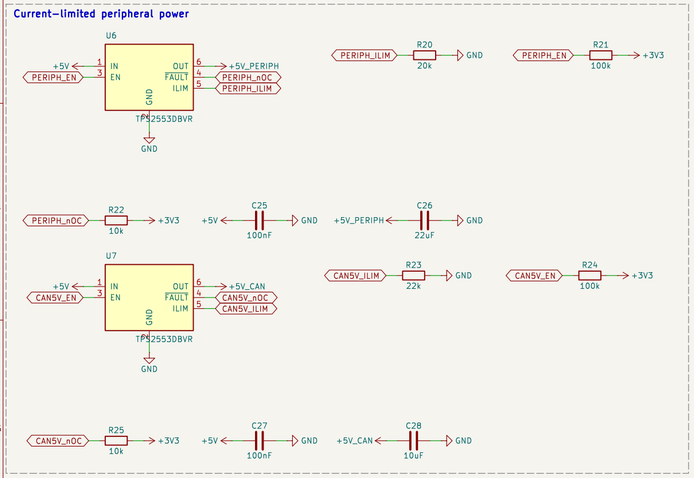
+3V3: TLV62569 2 A buck



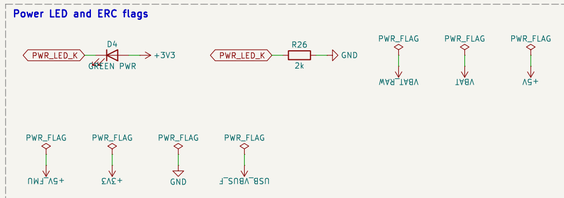
+3V3_SENS: low-noise LDO, MCU-switchable



Current-limited peripheral power



Power LED and ERC flags



Power tree: VBAT (2S–6S) -> RPP P-FET -> TVS -> TPS54360B buck (+5V, 3.5 A, 597 kHz).
+5V -> LM66100 ideal diode -> +5V_FMU (USB backs it up through a second LM66100 that the buck overrides).
+5V_FMU -> TLV62569 -> +3V3 (MCU, SD, CAN VIO); +5V_FMU -> LP5907 -> +3V3_SENS (IMUs, baro; MCU-switchable).
+5V -> TPS2553 x2 -> +5V_PERIPH (UARTs, 1.2 A) and +5V_CAN (1.0 A): a shorted cable trips its own switch.
Compensation values are checked in sim/buck (ngspice).

Generated from design.py; edit the source, not this file
Pin map: PX4 FMUv6C (Pixhawk 6C) derivative, STM32H743VIT6 LQFP-100
DRAFT UNTIL INDEPENDENT REVIEW. Not for fabrication.

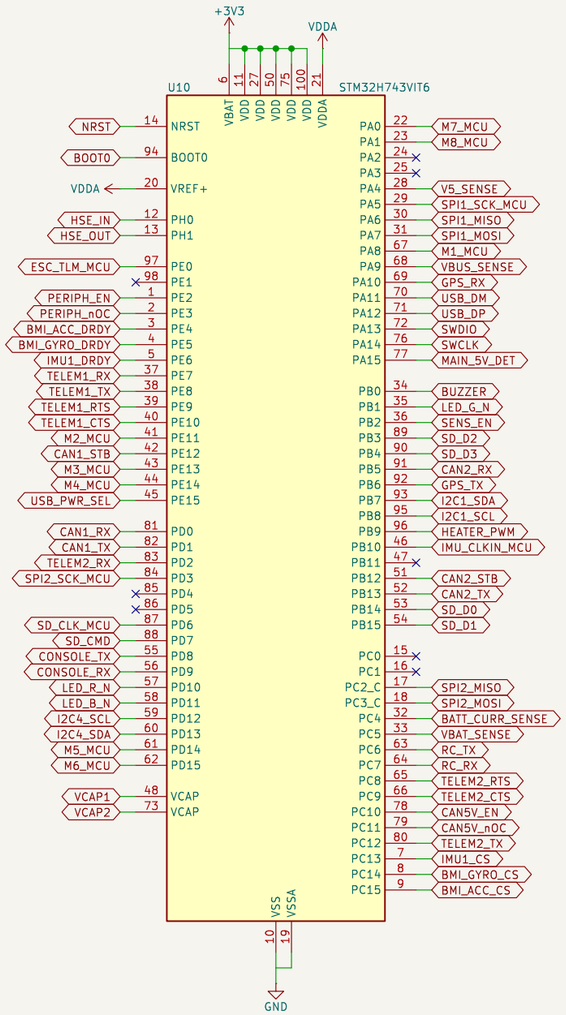
Daniel K (dvntekim)

Sheet: /Power/
File: power.kicad_sch

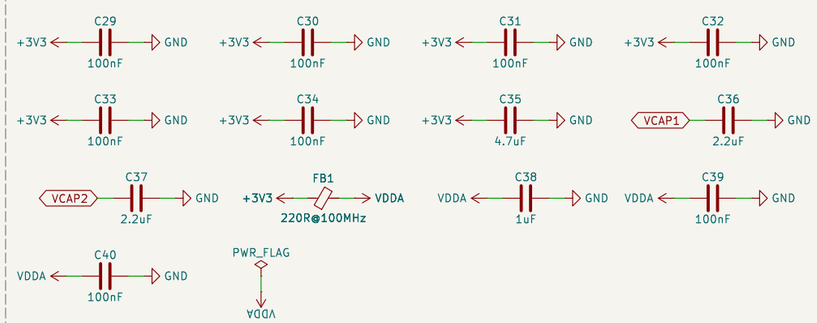
Title: GPS-denied quadrotor flight controller: Power

Size: A2	Date: 2026-11-07	**Rev: B**
KiCad E.D.A. 9.0.8		Id: 2/7

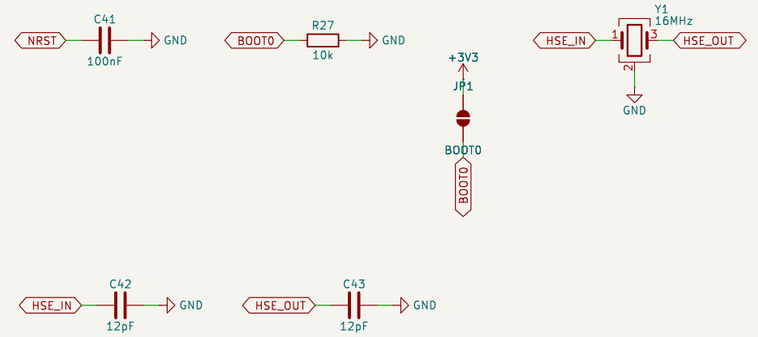
STM32H743VIT6



Decoupling (AN4938)



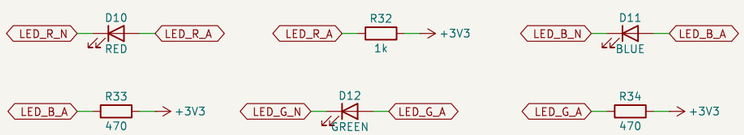
Reset, boot, clock



Series resistors at MCU



Status LEDs



STM32H743VIT6. Pin map follows PX4 FMUv6C; see pinmap.md for every pin's alternate-function check.
Decoupling per ST AN4938: 100 nF per VDD pin + 4.7 uF bulk, 2.2 uF per VCAP, VDDA through a ferrite.
BOOT0 bridge enters ST ROM DFU on a blank chip. SCK/CLK series resistors sit at the MCU.

Generated from design.py; edit the source, not this file
Pin map: PX4 FMUv6C (Pixhawk 6C) derivative, STM32H743VIT6 LQFP-100
DRAFT UNTIL INDEPENDENT REVIEW. Not for fabrication.

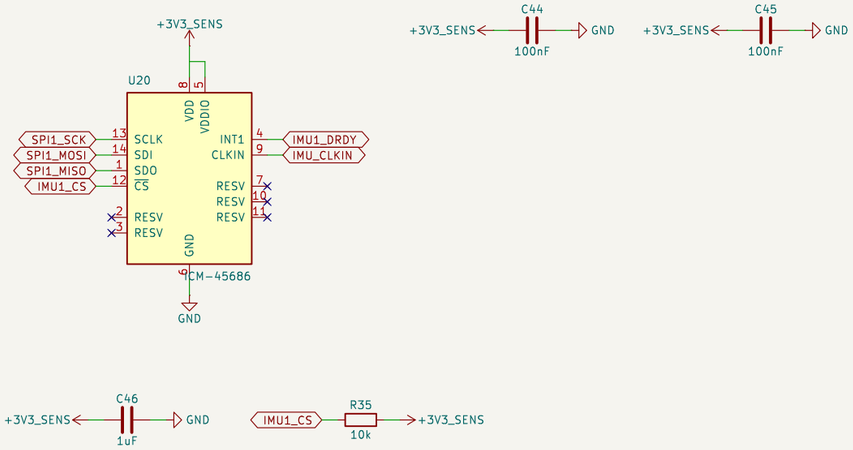
Daniel K (dvntekim)

Sheet: /MCU/
File: mcu.kicad_sch

Title: GPS-denied quadrotor flight controller: MCU

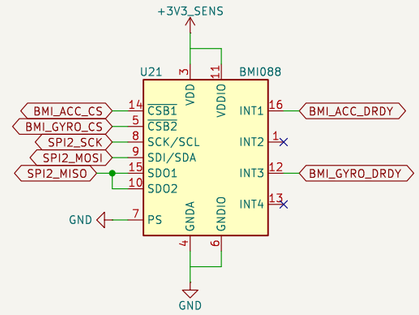
Size: A3	Date: 2026-11-07	**Rev: B**
KiCad E.D.A. 9.0.8		Id: 3/7

IMU 1: ICM-45686 on SPI1

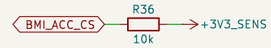


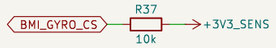


IMU 2: BMI088 on SPI2 (rev B: stays on the main board, unheated; see docs/revB_thermal.md)

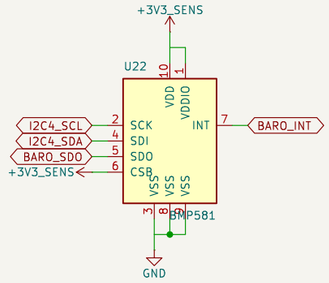


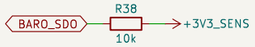


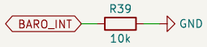


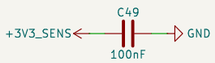


Barometer: BMP581 on I2C4

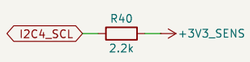






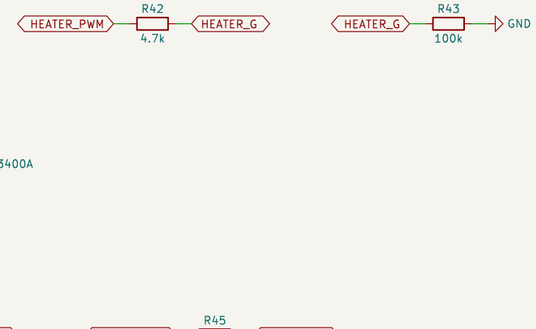


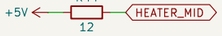


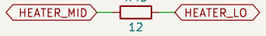




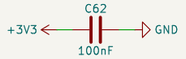
IMU heater (rev B: on the isolated IMU island)

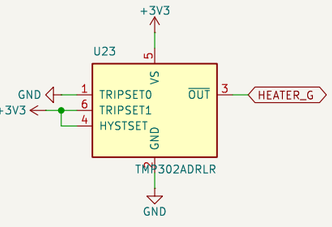






IMU heater hardware cut-off (rev B)





ICM-45686 on SPI1 with a 32 kHz CLKIN from TIM2_CH3 (PX4: icm45686 -C 32000), BMI088 on SPI2 (different vendor, own bus).
BMP581 on I2C4 at 0x47. All sensors on +3V3_SENS, which the MCU can power-cycle.
Heater: 4 x 120R around the IMUs, AO3400A on TIM17_CH1 (PX4 heater driver holds IMU temperature).

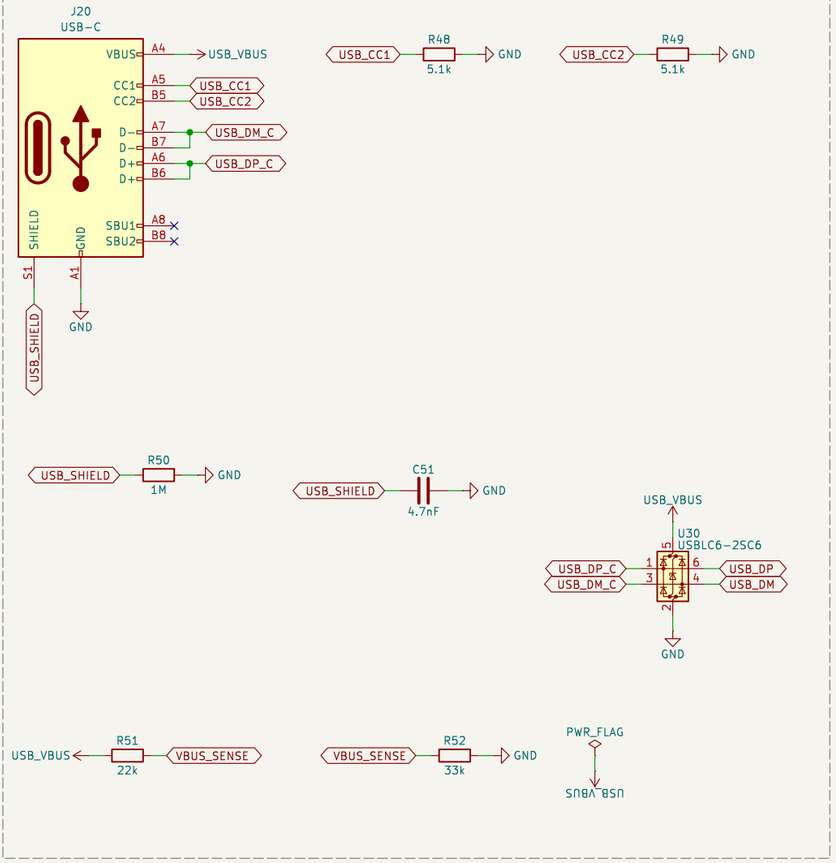
Generated from design.py; edit the source, not this file
Pin map: PX4 FMUv6C (Pixhawk 6C) derivative, STM32H743VIT6 LQFP-100
DRAFT UNTIL INDEPENDENT REVIEW. Not for fabrication.
Daniel K (dvntekim)

Sheet: /Sensors/
File: sensors.kicad_sch

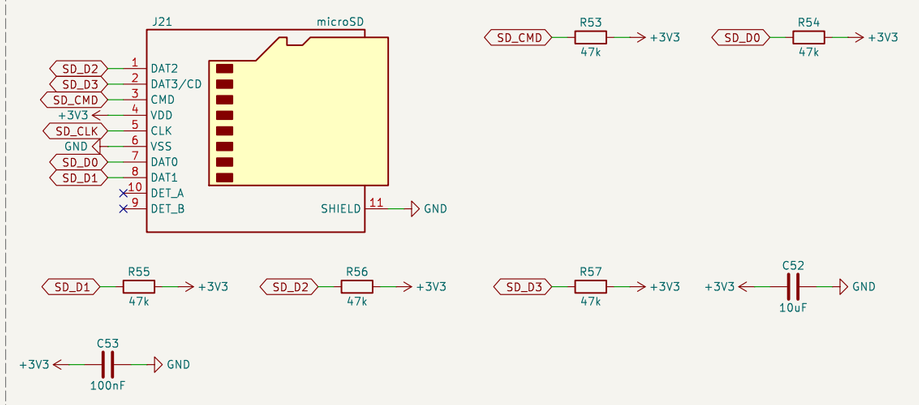
Title: GPS-denied quadrotor flight controller: Sensors

Size: A3	Date: 2026-11-07	**Rev: B**
KiCad E.D.A. 9.0.8		Id: 4/7

USB-C device



microSD (SDMMC2, 4-bit)



microSD on SDMMC2 (4-bit). USB-C device with 5.1k Rd, USBLC6 ESD, VBUS sense to PA9.

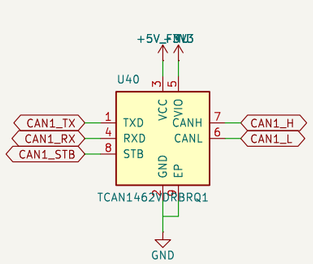
Generated from design.py; edit the source, not this file
Pin map: PX4 FMUv6C (Pixhawk 6C) derivative, STM32H743VIT6 LQFP-100
DRAFT UNTIL INDEPENDENT REVIEW. Not for fabrication.
Daniel K (dvntekim)

Sheet: /Storage + USB/
File: storage_usb.kicad_sch

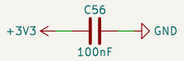
Title: GPS-denied quadrotor flight controller: Storage + USB

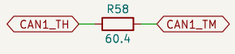
Size: A3	Date: 2026-11-07	**Rev: B**
KiCad E.D.A. 9.0.8		Id: 5/7

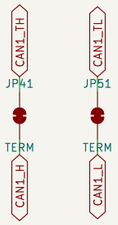
CAN1: TCAN1462V, split termination

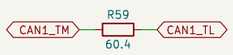






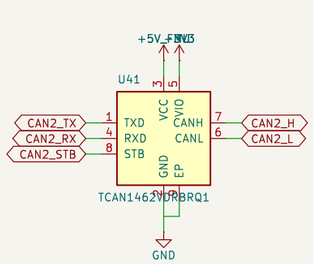




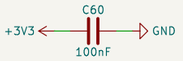


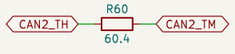


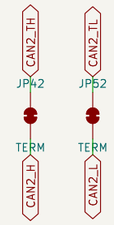
CAN2: TCAN1462V, split termination

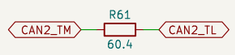


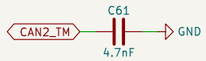












Two TCAN1462V CAN FD transceivers (8 Mbit/s, signal improvement). CAN1: DroneCAN flow + ToF. CAN2: reserved for own ESC.
Split termination is bridged by default; cut JP4x/JP5x when the FC is not a bus end.

Generated from design.py; edit the source, not this file
Pin map: PX4 FMUv6C (Pixhawk 6C) derivative, STM32H743VIT6 LQFP-100
DRAFT UNTIL INDEPENDENT REVIEW. Not for fabrication.

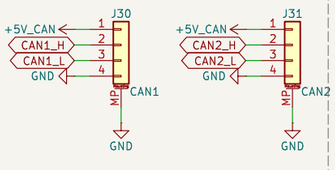
Daniel K (dvntekim)

Sheet: /CAN/
File: can.kicad_sch

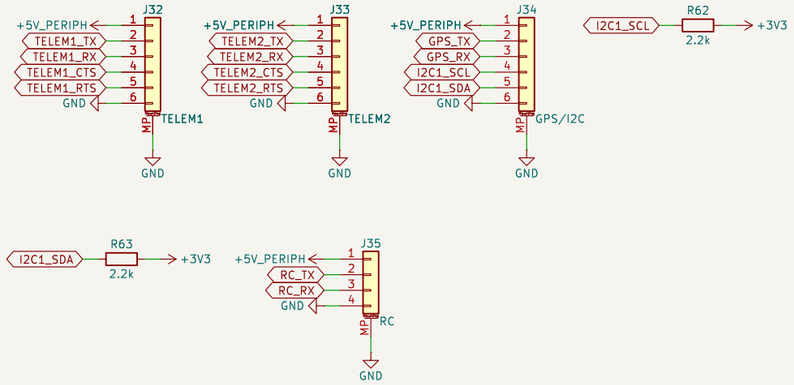
Title: GPS-denied quadrotor flight controller: CAN

Size: A3	Date: 2026-11-07	**Rev: B**
KiCad E.D.A. 9.0.8		Id: 6/7

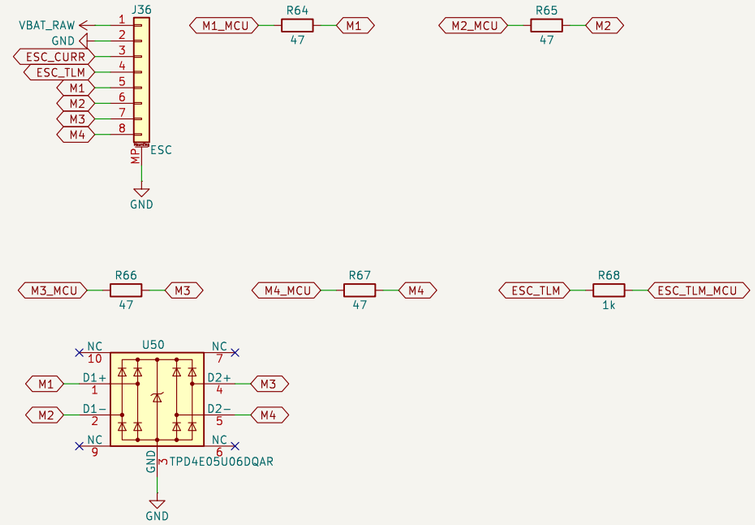
CAN connectors



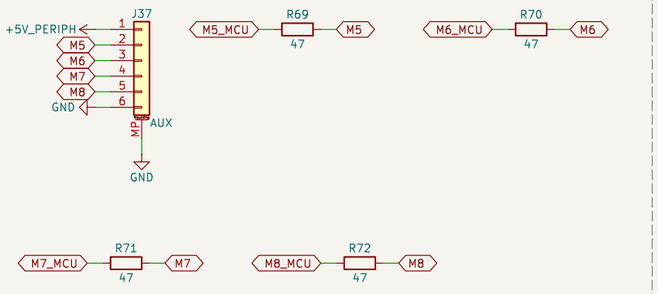
Serial ports



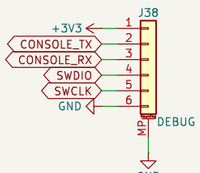
ESC harness (BDShot M1–M4, telemetry, current)



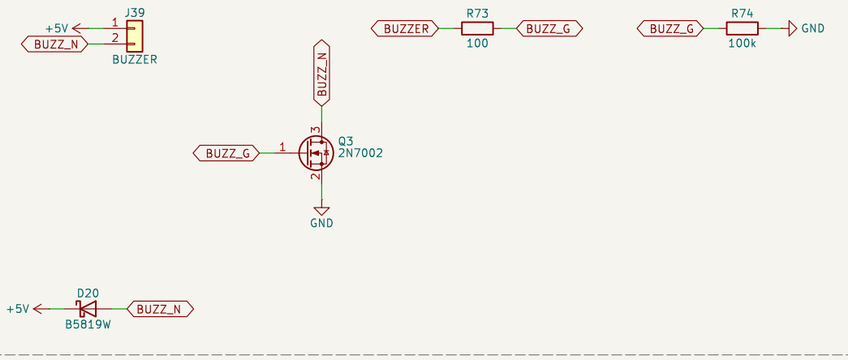
AUX: M5–M8, camera trigger/capture



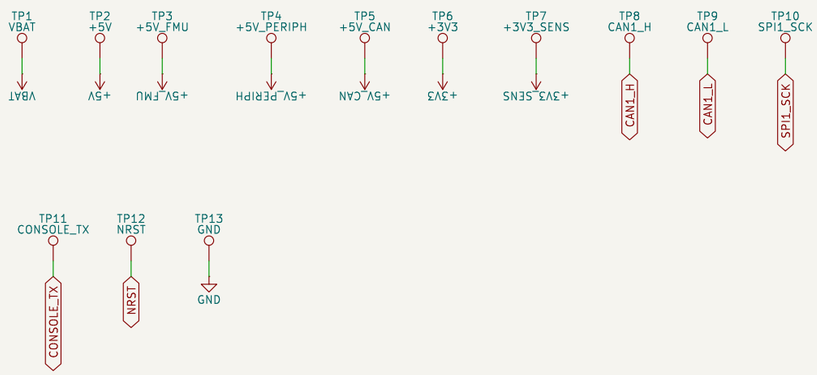
Debug (Pixhawk Debug Mini)



Buzzer



Test points



Mechanical



Pixhawk-standard JST-GH for peripherals, JST-SH for ESC and debug (DS-009).
AUX: M7 = camera trigger, M8 = camera capture for hardware time sync with a VIO camera.

Generated from design.py; edit the source, not this file
Pin map: PX4 FMUv6C (Pixhawk 6C) derivative, STM32H743VIT6 LQFP-100
DRAFT UNTIL INDEPENDENT REVIEW. Not for fabrication.

Daniel K (dvntekim)

Sheet: /Connectors/
File: connectors.kicad_sch

Title: GPS-denied quadrotor flight controller: Connectors

Size: A3	Date: 2026-11-07	**Rev: B**
KiCad E.D.A. 9.0.8		Id: 7/7